BeeMonitor

Edward Amoah
Oct 13, 15, 20, 22

Monitor foraging and nesting activity of cavity nesting bees

(A)

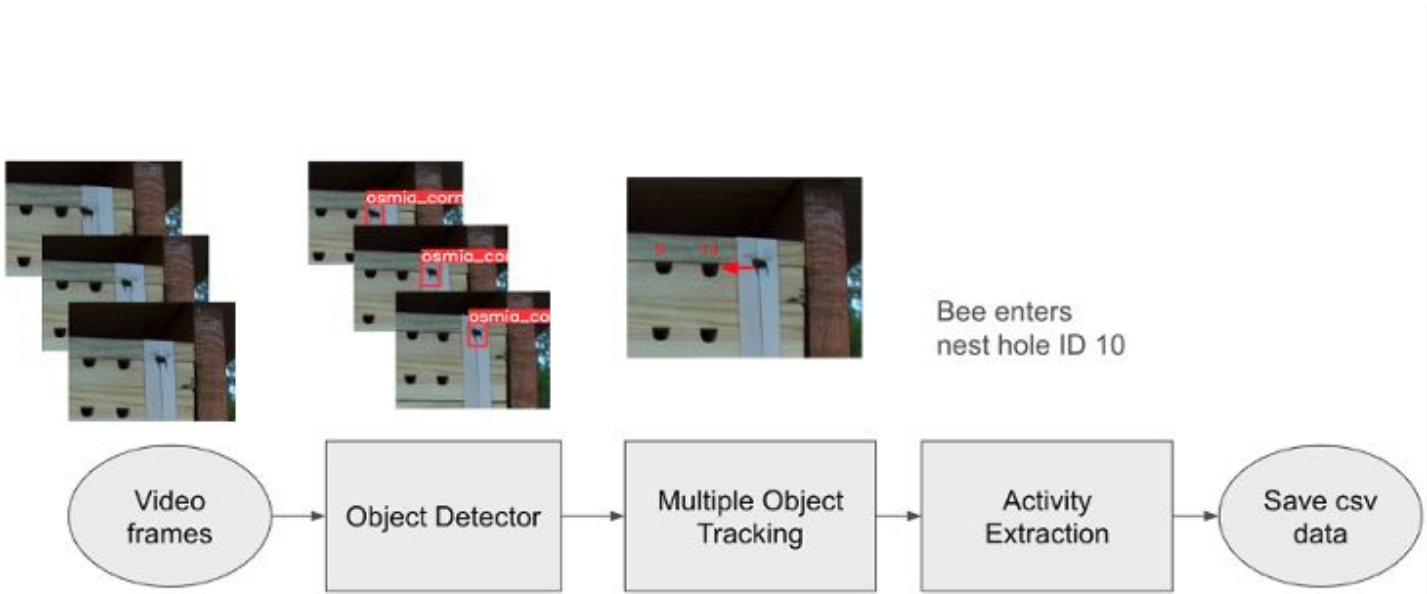


(B)





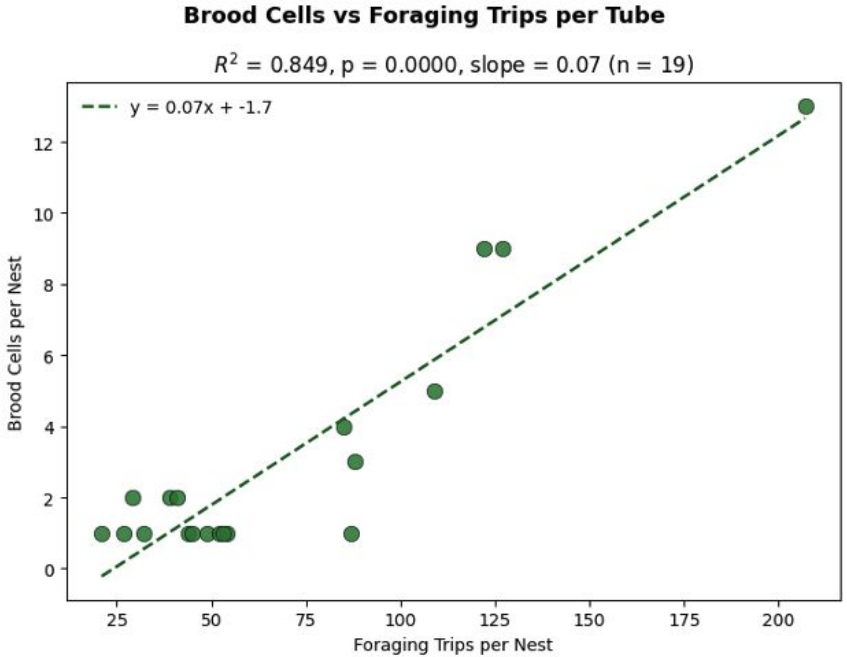
BeeMonitor computer vision analysis pipeline



Annotated video from BeeMonitor analysis



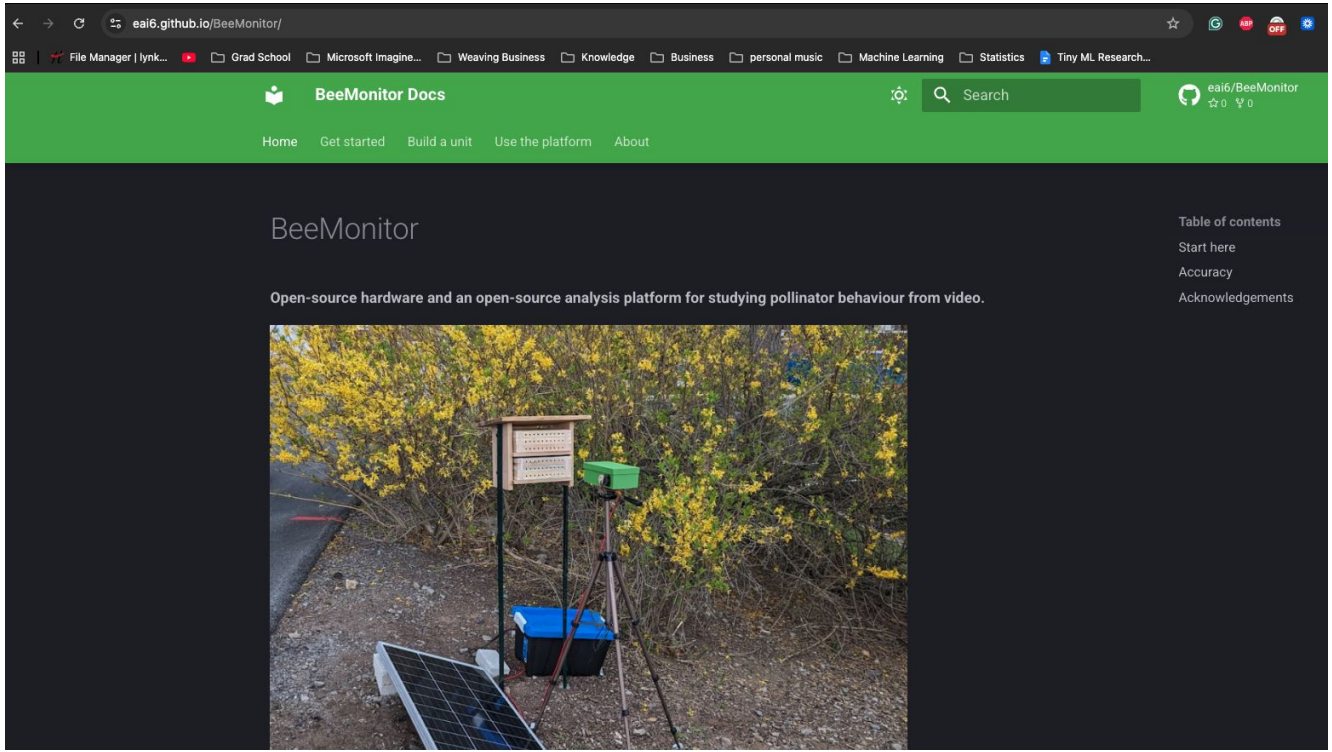
BeeMonitor detected foraging trips are correlated with brood cell number per nest-tube



Program Outline

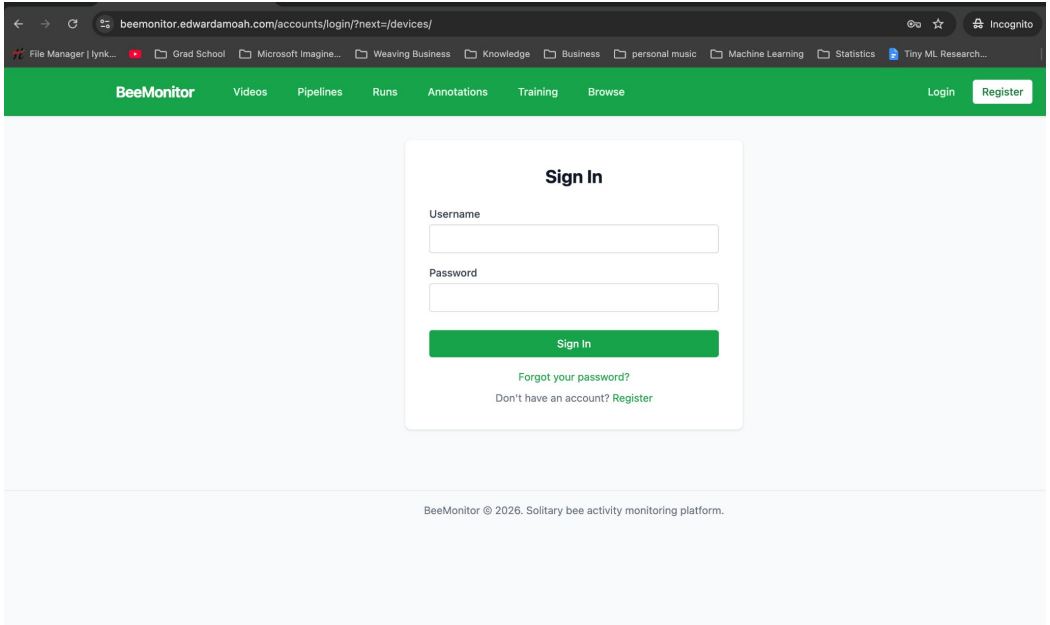
1. Hands-on assembly and deployment of BeeMonitor hardware
2. Hands-on annotations and programing of CV pipelines on platform
3. Review results of analysis from fine-tuned detector

Docs and additional materials are open-source on github



https://eai6.github.io/BeeMonitor/

Create an account on the BeeMonitor platform to practice camera deployment



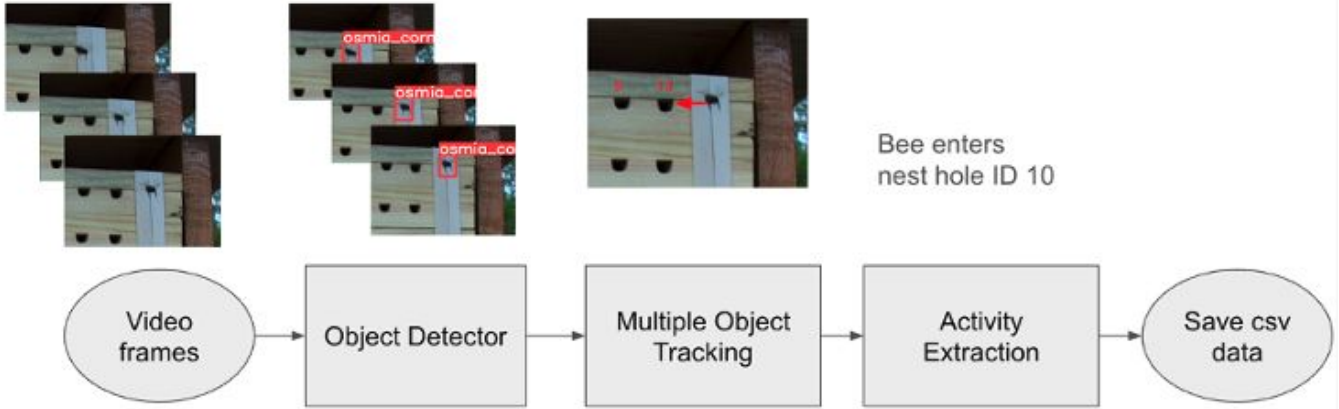
https://beemonitor.edwardamoah.com/

Class discussion

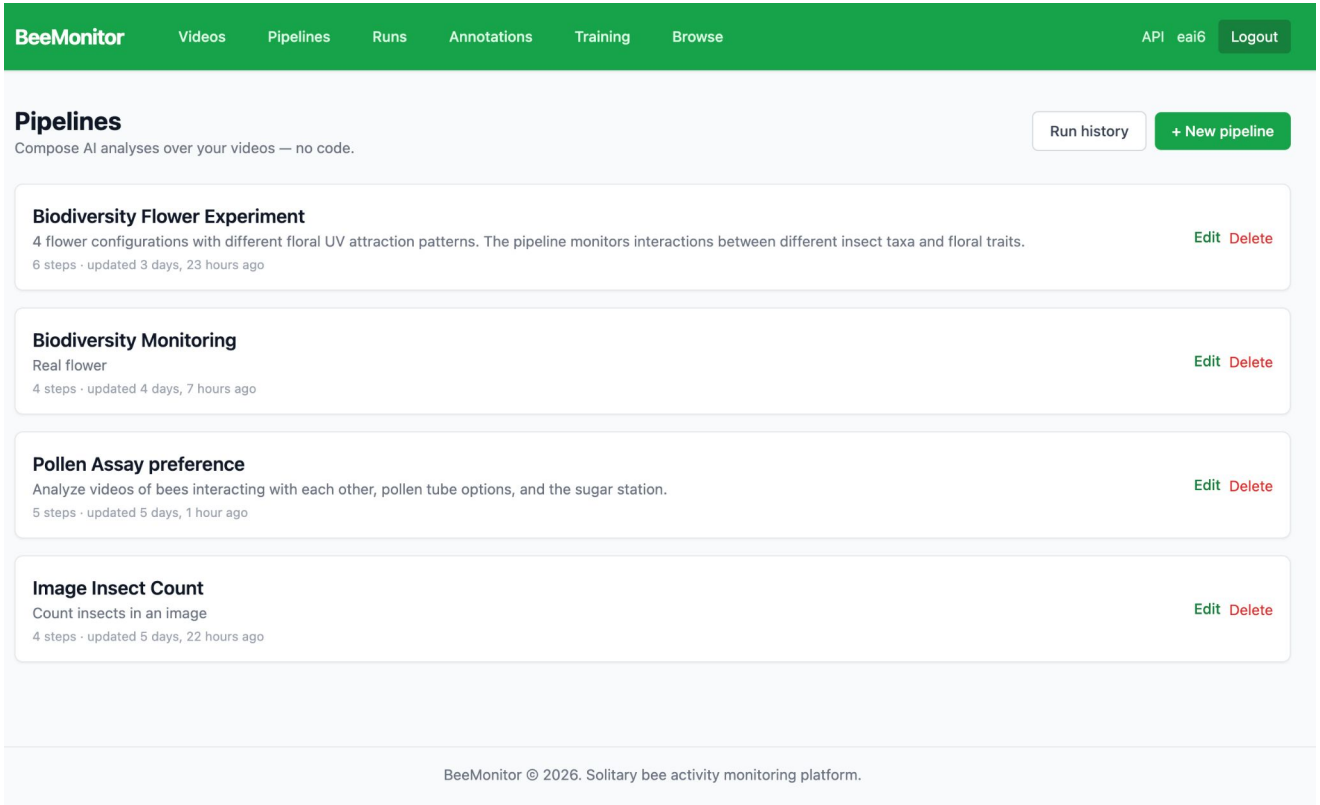
- Share the name of your BeeMonitor
- Share with the class how you hope to use automated monitoring for your research

Video Data Analysis

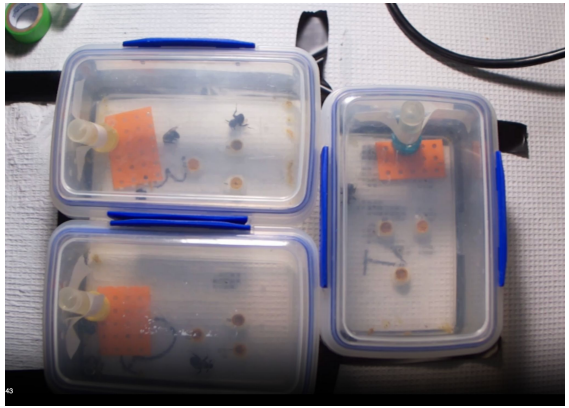
1. Drag and drop computer vision pipeline programing
2. Annotation and detector fine-tuning

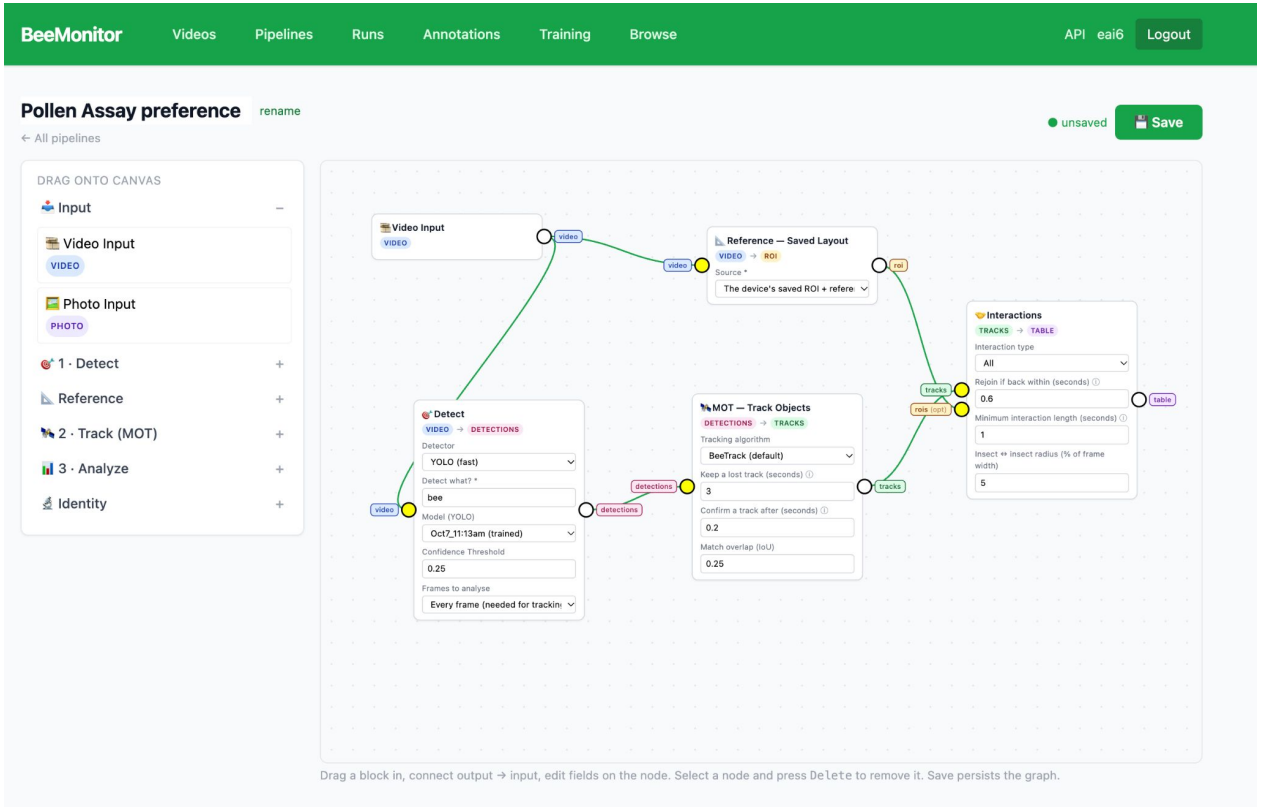


Computer vision pipelines

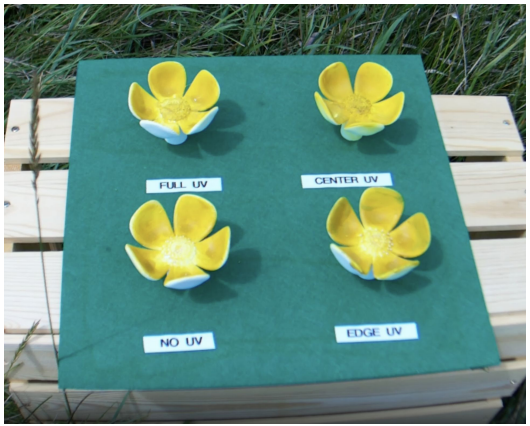


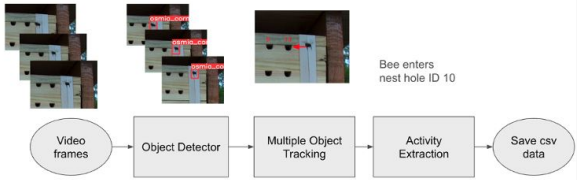
Pollen assay preference pipeline





Biodiversity flower experiment analysis pipeline





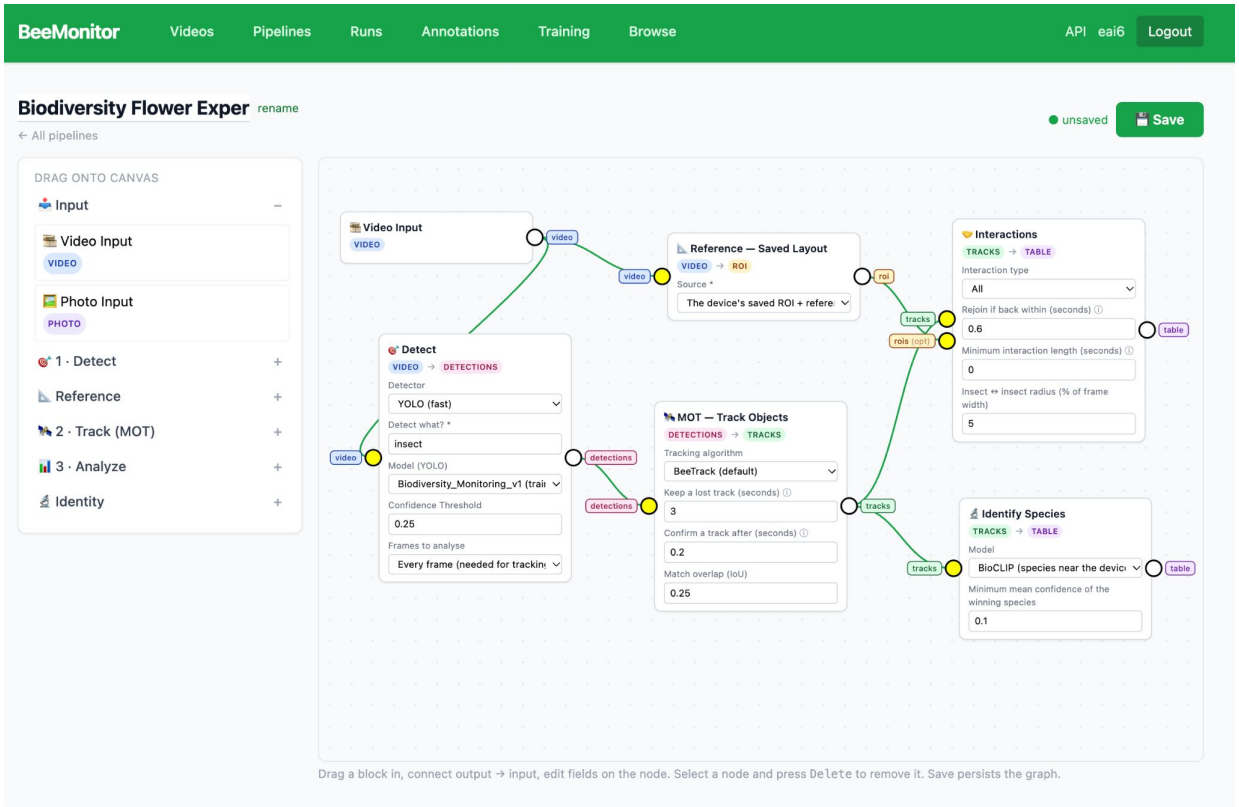
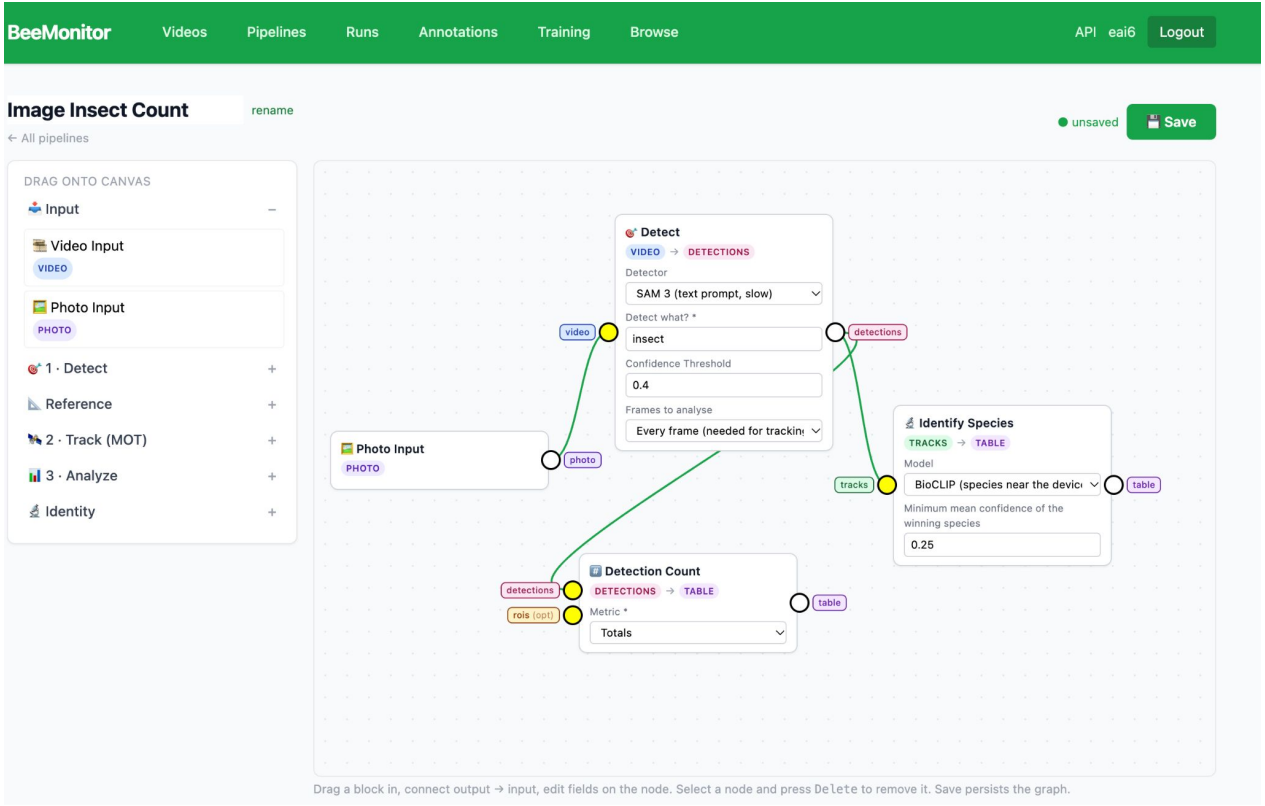
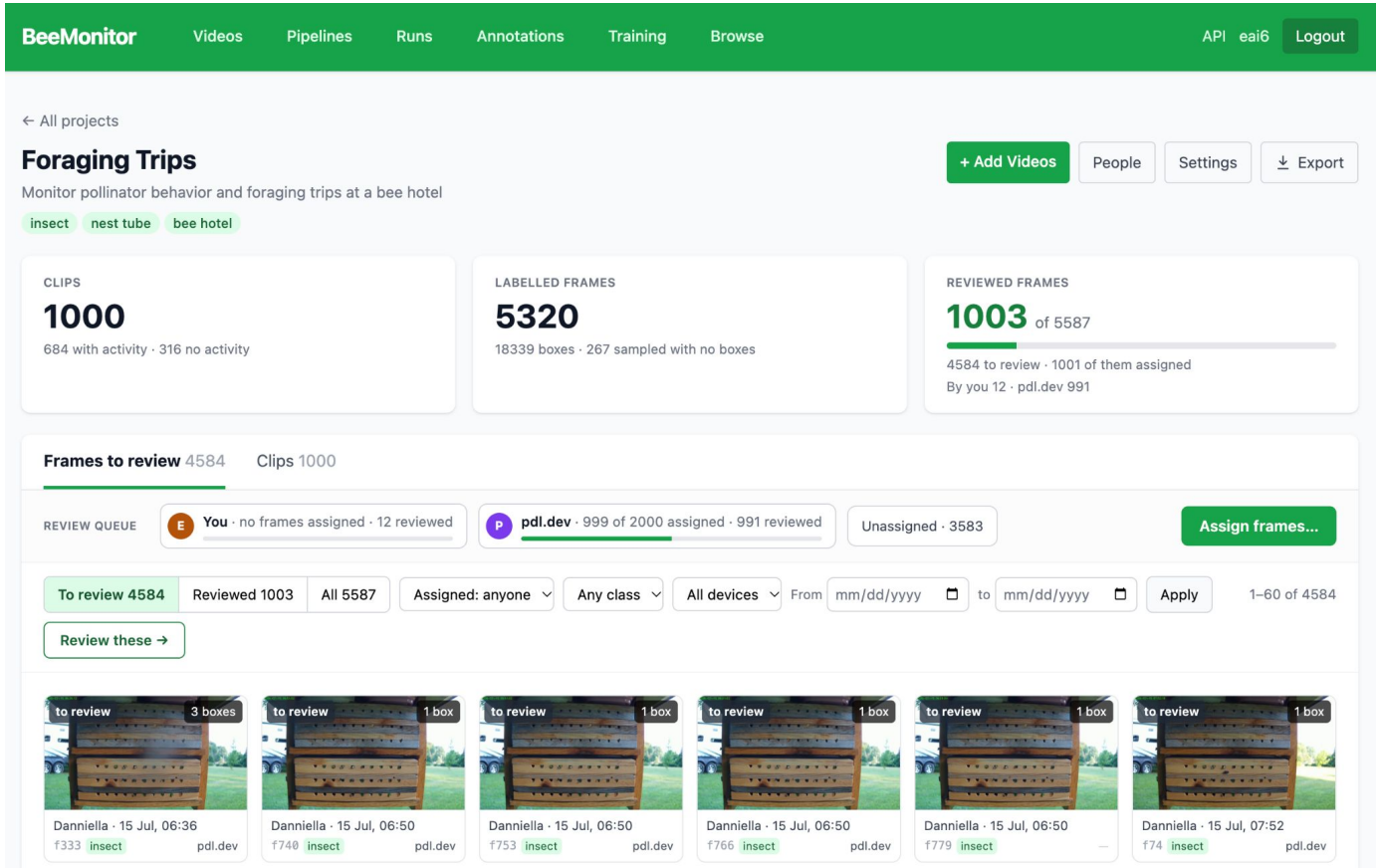


Image insect count

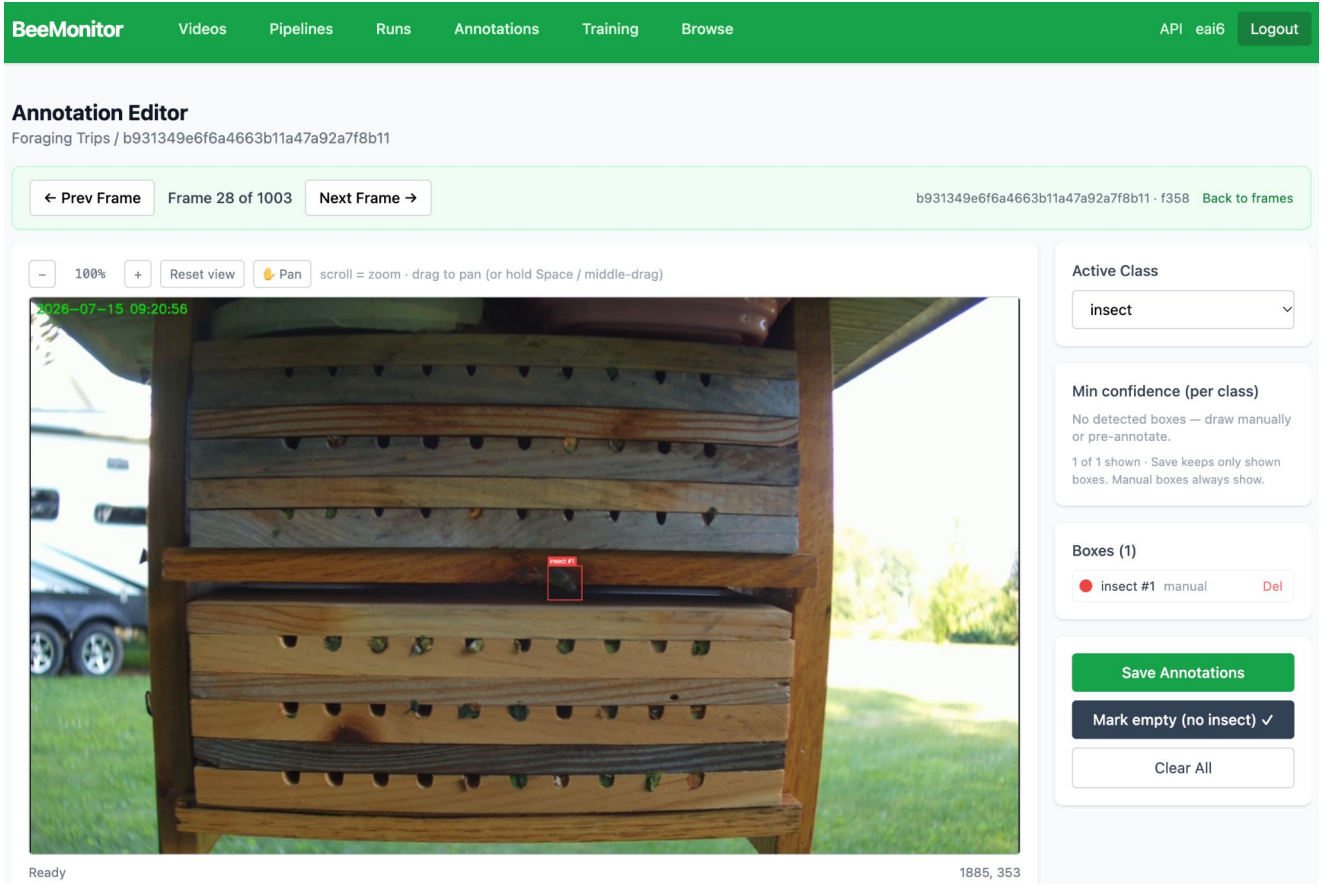




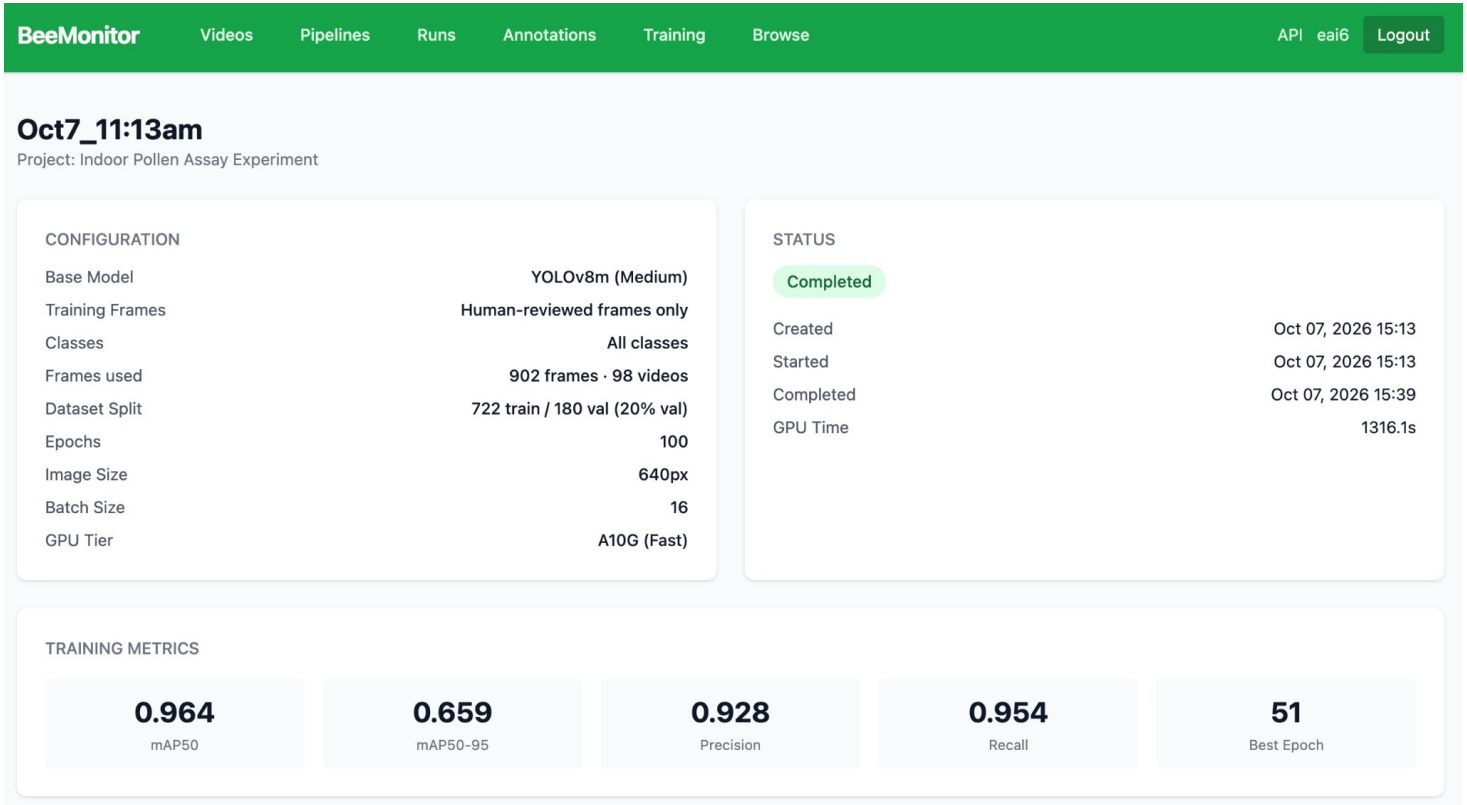
Annotation


BeeMonitor
Videos
Pipelines
Runs
Annotations
Training
Browse
API
eai6
Logout
← All projects
Foraging Trips
Monitor pollinator behavior and foraging trips at a bee hotel
insect
nest tube
bee hotel
+ Add Videos
People
Settings
Export
CLIPS
1000
684 with activity · 316 no activity
LABELLED FRAMES
5320
18339 boxes · 267 sampled with no boxes
REVIEWED FRAMES
1003 of 5587
4584 to review · 1001 of them assigned
By you 12 · pdl.dev 991
Frames to review 4584
Clips 1000
REVIEW QUEUE
You · no frames assigned · 12 reviewed
pdl.dev · 999 of 2000 assigned · 991 reviewed
Unassigned · 3583
Assign frames...
To review 4584
Reviewed 1003
All 5587
Assigned: anyone
Any class
All devices
From
mm/dd/yyyy
to
mm/dd/yyyy
Apply
1–60 of 4584
Review these →
to review
3 boxes
1 box
Danniella · 15 Jul, 06:36
Danniella · 15 Jul, 06:50
Danniella · 15 Jul, 07:52
insect
pdl.dev

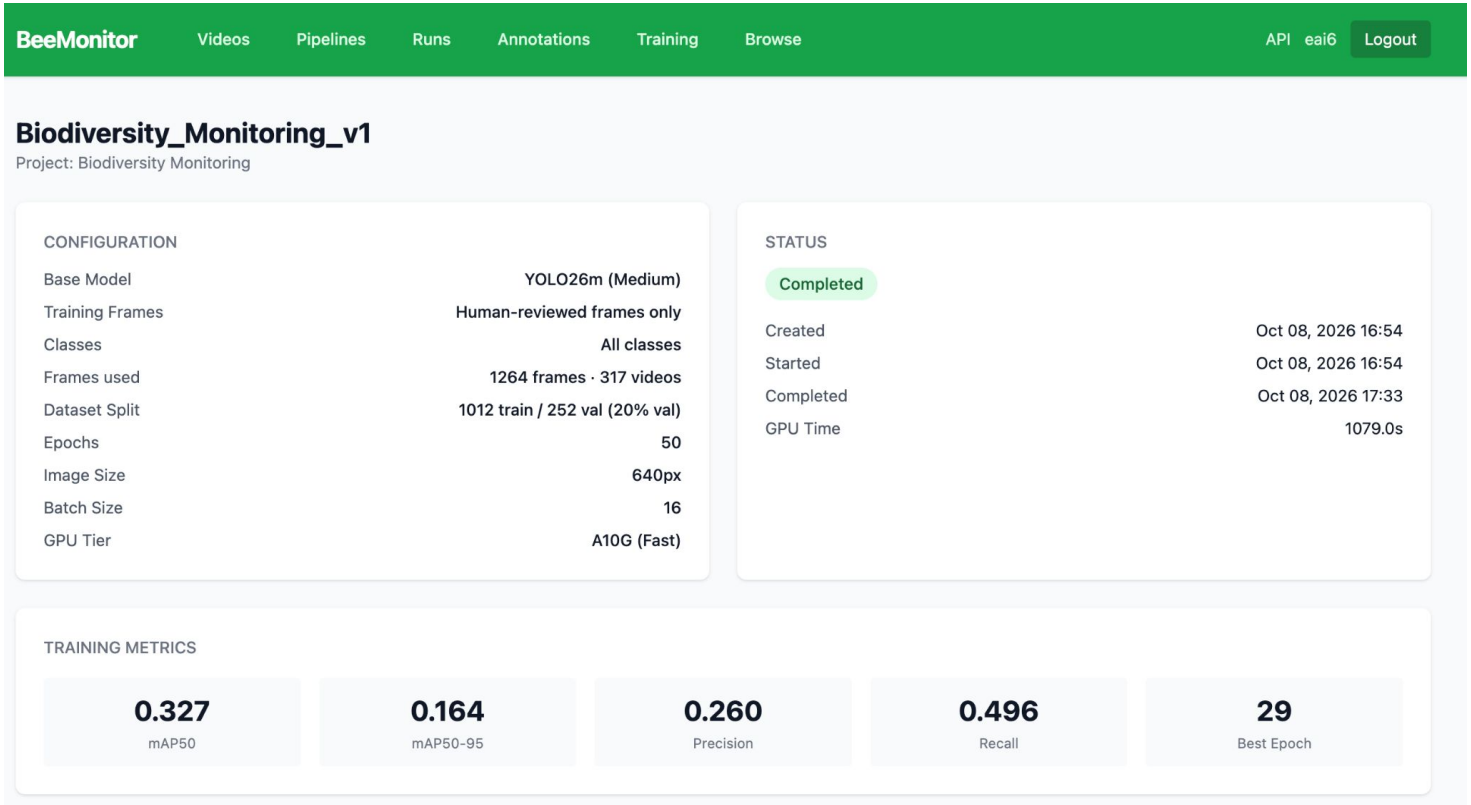
Annotation


BeeMonitor
Videos
Pipelines
Runs
Annotations
Training
Browse
API
eai6
Logout
Annotation Editor
Foraging Trips / b931349e6f6a4663b11a47a92a7f8b11
← Prev Frame
Frame 28 of 1003
Next Frame →
b931349e6f6a4663b11a47a92a7f8b11 · f358
Back to frames
100%
Reset view
Pan
scroll = zoom · drag to pan (or hold Space / middle-drag)
2026-07-15 09:20:56
Active Class
insect
Min confidence (per class)
No detected boxes — draw manually or pre-annotate.
1 of 1 shown · Save keeps only shown boxes. Manual boxes always show.
Boxes (1)
insect #1 manual
Del
Save Annotations
Mark empty (no insect) ✓
Clear All
Ready
1885, 353

Training


BeeMonitor
Videos
Pipelines
Runs
Annotations
Training
Browse
API
eai6
Logout
Oct7_11:13am
Project: Indoor Pollen Assay Experiment
CONFIGURATION
Base Model
YOLOv8m (Medium)
Training Frames
Human-reviewed frames only
Classes
All classes
Frames used
902 frames · 98 videos
Dataset Split
722 train / 180 val (20% val)
Epochs
100
Image Size
640px
Batch Size
16
GPU Tier
A10G (Fast)
STATUS
Completed
Created
Oct 07, 2026 15:13
Started
Oct 07, 2026 15:13
Completed
Oct 07, 2026 15:39
GPU Time
1316.1s
TRAINING METRICS
0.964
mAP50
0.659
mAP50-95
0.928
Precision
0.954
Recall
51
Best Epoch

Training


BeeMonitor
Videos
Pipelines
Runs
Annotations
Training
Browse
API
eai6
Logout
Biodiversity_Monitoring_v1
Project: Biodiversity Monitoring
CONFIGURATION
Base Model
YOLO26m (Medium)
Training Frames
Human-reviewed frames only
Classes
All classes
Frames used
1264 frames · 317 videos
Dataset Split
1012 train / 252 val (20% val)
Epochs
50
Image Size
640px
Batch Size
16
GPU Tier
A10G (Fast)
STATUS
Completed
Created
Oct 08, 2026 16:54
Started
Oct 08, 2026 16:54
Completed
Oct 08, 2026 17:33
GPU Time
1079.0s
TRAINING METRICS
0.327
mAP50
0.164
mAP50-95
0.260
Precision
0.496
Recall
29
Best Epoch

In class activity

Build a pipeline for foraging activity analysis

https://forms.gle/uqUJPoSiryMcUtY97

